Mohak A. Prakash

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Education

University of California, San Diego — Expected June 2027

- **B.S. Double Major in Data Science & Cognitive Science (Neuroscience)**, GPA: 3.816/4.00, Provost Honors
- Relevant Coursework: Probabilistic Modeling & Machine Learning, Data Mining, Signal Processing, Systems for Scalable Analytics, Data Management, Reinforcement Learning, IoT Sensors & Embedded Systems

Skills

Python, JavaScript, TypeScript, React, Flutter, FastAPI, Flask, Pandas, TensorFlow, PyTorch, Scikit-learn, Cvxpy, OpenCV, MediaPipe, LangChain, LangGraph, LlamaIndex, SQL, MongoDB, Firebase, AWS, GCP, Docker, Kubernetes, Spark, Dask, PCA, FFT/Signal Processing, Clustering (K-means, DBSCAN, GMM), SVM, Random Forests, Q-Learning, Transformers, OCR, LLMs & VLMs (OpenAI, Claude), RAG, Knowledge Graphs, NER, RelExtract, MCP, Twilio, Ngrok

Work Experience

Applied AI Intern — *BCG (Boston Consulting Group) X, San Francisco* — Jun 2026 - Present

- Building GenAI product prototypes and internal AI tooling for enterprise client engagements.

Research Assistant — *Knight Lab, University of California San Diego* — Jan 2025 - Present

- Scraped and cleaned 2,000+ research articles across 50+ journals at a 90% success rate for the MicrobiologyMetadataCrisis Project (250+ members), across all MMC1 and MMC2 projects.
- Built the relation-extraction pipeline on Qwen3.5-Opus4.6-Distilled (GGUF), reaching 0.77 F1 on a locked 15-paper gold-standard test set run on UCSD's DSMLP cluster (1 H100 GPU).
- Rebuilt the evaluation harness to score predicted microbe-disease pairs against ground truth with Hungarian matching (scipy) and NCBI Taxonomy normalization (ete3) for stable, reproducible F1 across model iterations.

Research Assistant — *Proto Lab, UC San Diego Design Lab* — Sep 2025 - Jan 2026

- Built AI-powered meeting simulator enabling multi-round agent discussions with configurable aggressivity; reduced latency from 12-14s to 0.1-0.5s using custom sliding-window optimization algorithm.
- Developed speaker diarization and persona generation pipeline using Pyannote, Rev AI, NVIDIA NeMo/Riva to extract real speech patterns for human-like AI persona creation.

Data Analytics Intern — *EXL Services* — Jun 2025 - Sep 2025

- Built RiskXpert (renamed), an agentic RAG tool that ingested meeting minutes and enterprise documents to automate risk assessment and control identification, streamlining underwriting and compliance workflows.
- Benchmarked OCR pipelines (AWS Textract, Google DocAI, PaddleOCR + LayoutLMv3, and a custom CNN+RNN+CRF) across varied document layouts to select the best fit extractor.

Projects

Winnow | *Modal, FastAPI, PyTorch, Qwen, Next.js, Deepgram* — UC Berkeley AI '26

- Built a prompt-compression engine that runs token-space (LLMLingua-2, AttentionRAG) and model-space (TurboQuant 4-bit KV-cache) compressors in parallel, reconciling them with an order-preserving char-span merge to cut context cost 3-4x while matching FP16 output.
- Shipped a live voice-sample compression demo (Deepgram transcription piped straight into the compressor) and a model playground exposing our own compressed models alongside Claude and GPT for side-by-side comparison.

Few Word Do Trick | *LangGraph, LangChain, GPT-4o, Whisper, scikit-learn, Firestore* — Winner @ CalHacks '25

- Built an AAC platform for people with speech impediments: extracts spectral EEG features (FFT, PSD, PCA, ICA) from a Muse headset and classifies binary emotional state with a Random Forest at 91% test / 95% train accuracy.
- Orchestrated context management with LangGraph and LangChain over Firestore, turning a few typed keywords plus the detected emotion into context-aware, emotionally graded speech.

Clueless | *Claude, Neo4j, Pinecone, Cerebras, React/TypeScript, Plasmo* — Winner @ HackMIT '25

- Developed an AI web-navigation assistant that auto-maps any web app into a Neo4j knowledge graph and runs BFS traversal over it to deliver real-time, step-by-step voice and text UI guidance.
- Served through a Plasmo browser extension with Cerebras-hosted inference and Pinecone semantic retrieval over UI elements for low-latency responses.

TaxDaddy | *Twilio, WebSockets, Pinecone RAG, MongoDB* — Winner @ Hacklytics '25

- Built a real-time voice tax assistant that answers filing questions over a live phone call, streaming audio through Twilio and WebSockets against a Pinecone RAG backend over tax-code and IRS documents.
- Auto-populates tax forms from scanned documents and flags numerical irregularities with Benford's Law anomaly detection, with sessions persisted in MongoDB.

Achievements and Activities

7x Hackathon Winner: Voice Cursor Hackathon • CalHacks '25 • HealthLink '25 • HackMIT '25 • Hacklytics '25 • CodeDay '23 • YTBC'23 Top 50/12,000+

HDSI Department Tutor (DSC 40A, DSC 20) • **Triton NeuroTech Project Lead** • **DS3 DataHacks Organizer**